

Phase 2 — scaling with N_fe

Mercury (Z=80), all-electron, corrected SPARC-atomSFE. Fixed: p=20, q=45, domain=13 Bohr, omega=4, polynomial mesh (concentration 2). Channels = Lmax + 4, since Hg has 4f occupied (l_occ_max = 3). Lmax held at 30, so only n_quad moves. Exponents are local, between consecutive rows, against log(n_quad).

self-consistent RPA — scaling with N_fe

N_fe	n_quad	mem (MiB)	wall (s)	s/channel	mem exp	time exp
10	450	263.0	17.8	0.52	-	-
15	675	374.7	45.7	1.34	0.87	2.32
25	1125	650.1	149.4	4.39	1.08	2.32
40	1800	1390.3	460.8	13.55	1.62	2.40

memory 194 + 3.685e-04 n_quad^2 MiB at Lmax=30 (max err 3.5%); the quadratic term is 86% of the peak at the anchor
1067 + 10.36 Lmax MiB at n_quad=1800 (max err 1.7%)
time wall = channels x 3.170e-07 n_quad^2.34 s (max err 1.6%)
cross-check at n_quad=1800: wall = 20.4 + 14.10 x channels, so everything outside the RPA is 4.4% of the total

non-self-consistent RPA@DFT — scaling with N_fe

N_fe	n_quad	mem (MiB)	wall (s)	s/channel	mem exp	time exp
10	450	240.6	6.6	0.19	-	-
15	675	333.0	13.2	0.39	0.80	1.70
25	1125	586.0	40.5	1.19	1.11	2.19
40	1800	1193.9	129.3	3.80	1.51	2.47

memory 186 + 3.119e-04 n_quad^2 MiB at Lmax=30 (max err 3.5%); the quadratic term is 85% of the peak at the anchor
859 + 11.31 Lmax MiB at n_quad=1800 (max err 0.8%)
time wall = channels x 3.399e-07 n_quad^2.16 s (max err 9.7%)
cross-check at n_quad=1800: wall = +1.1 + 3.77 x channels, so everything outside the RPA is 0.9% of the total

Phase 2 — scaling with Lmax

Mercury (Z=80), all-electron, corrected SPARC-atomSFE. Fixed: p=20, q=45, domain=13 Bohr, omega=4, polynomial mesh (concentration 2). Channels = Lmax + 4, since Hg has 4f occupied (l_occ_max = 3). N_fe held at 40, so n_quad is constant at 1800 and only the channel count moves. Exponents are against log(channels).

self-consistent RPA — scaling with Lmax

Lmax	channels	mem (MiB)	wall (s)	s/channel	mem exp	time exp
5	9	1138.2	107.2	11.92	-	-
10	14	1153.6	177.7	12.70	0.03	1.14
20	24	1258.3	314.4	13.10	0.16	1.06
30	34	1390.3	460.8	13.55	0.29	1.10

memory 194 + 3.685e-04 n_quad^2 MiB at Lmax=30 (max err 3.5%); the quadratic term is 86% of the peak at the anchor
1067 + 10.36 Lmax MiB at n_quad=1800 (max err 1.7%)
time wall = channels x 3.170e-07 n_quad^2.34 s (max err 1.6%)
cross-check at n_quad=1800: wall = 20.4 + 14.10 x channels, so everything outside the RPA is 4.4% of the total

non-self-consistent RPA@DFT — scaling with Lmax

Lmax	channels	mem (MiB)	wall (s)	s/channel	mem exp	time exp
5	9	916.3	35.0	3.89	-	-
10	14	966.5	54.0	3.86	0.12	0.98
20	24	1093.9	91.4	3.81	0.23	0.98
30	34	1193.9	129.3	3.80	0.25	1.00

memory 186 + 3.119e-04 n_quad^2 MiB at Lmax=30 (max err 3.5%); the quadratic term is 85% of the peak at the anchor
859 + 11.31 Lmax MiB at n_quad=1800 (max err 0.8%)
time wall = channels x 3.399e-07 n_quad^2.16 s (max err 9.7%)
cross-check at n_quad=1800: wall = +1.1 + 3.77 x channels, so everything outside the RPA is 0.9% of the total

Phase 2 — total-energy convergence

Mercury (Z=80), all-electron, corrected SPARC-atomSFE. Fixed: p=20, q=45, domain=13 Bohr, omega=4, polynomial mesh (concentration 2). Channels = Lmax + 4, since Hg has 4f occupied (l_occ_max = 3). Deviations are measured against a projected (Lmax-extrapolated) reference, not against the largest run.

self-consistent RPA — total-energy convergence

N_fe	Lmax	E (Ha)	E - E_ref (Ha)	within 1e-4	within 1e-5
40	5	-18409.776929186	4.64e-01	no	no
40	10	-18410.218848242	2.20e-02	no	no
40	20	-18410.240084032	7.51e-04	no	no
10	30	-18410.241904062	1.07e-03	no	no
15	30	-18410.241666979	8.32e-04	no	no
25	30	-18410.241629856	7.95e-04	no	no
40	30	-18410.240730760	1.04e-04	no	no

reference E_inf = -18410.240834913 Ha, from E(L) = E_inf + c L^-k on the top three Lmax points; k = 4.87
deviation at Lmax=30 is 1.04e-04 Ha; Lmax needed for 1e-4 is 30, for 1e-5 is 49
the four N_fe rows all sit at Lmax=30, so their deviation IS the Lmax error -- the N_fe sweep itself spans only 1.2e-03 Ha with non-monotone increments, i.e. N_fe is not the limiting parameter

non-self-consistent RPA@DFT — total-energy convergence

N_fe	Lmax	E (Ha)	E - E_ref (Ha)	within 1e-4	within 1e-5
40	5	-18413.162880405	4.69e-01	no	no
40	10	-18413.610108664	2.20e-02	no	no
40	20	-18413.631348768	7.18e-04	no	no
10	30	-18413.631984602	8.20e-05	yes	no
15	30	-18413.631983599	8.30e-05	yes	no
25	30	-18413.631981613	8.50e-05	yes	no
40	30	-18413.631969526	9.71e-05	yes	no

reference E_inf = -18413.632066577 Ha, from E(L) = E_inf + c L^-k on the top three Lmax points; k = 4.93
deviation at Lmax=30 is 9.71e-05 Ha; Lmax needed for 1e-4 is 30, for 1e-5 is 48
the four N_fe rows all sit at Lmax=30, so their deviation IS the Lmax error -- the N_fe sweep itself spans only 1.5e-05 Ha with non-monotone increments, i.e. N_fe is not the limiting parameter

Phase 2 — projections

Mercury (Z=80), all-electron, corrected SPARC-atomSFE. Fixed: p=20, q=45, domain=13 Bohr, omega=4, polynomial mesh (concentration 2). Channels = Lmax + 4, since Hg has 4f occupied (l_occ_max = 3). Targets are (N_fe=150, Lmax=80) and (N_fe=200, Lmax=100).

Projected memory and wall time

mode	N_fe	Lmax	n_quad	channels	mem low (GiB)	mem high (GiB)	s/channel	wall (h)
scRPA	150	80	6750	84	17.1	23.7	296.68	6.9
scRPA	200	100	9000	104	30.0	47.0	582.10	16.8
RPA@DFT	150	80	6750	84	14.6	21.8	60.88	1.4
RPA@DFT	200	100	9000	104	25.6	44.2	113.18	3.3

`mem low` carries the Lmax term as additive and n_quad-independent; `mem high` lets it grow with the n_quad^2 term. Only one n_quad was sampled in the angular sweep, so which is right is undetermined and the true wall time needs no such caveat: the channel count enters linearly, verified to 4.4% (scRPA) and 0.9% (RPA@DFT) at n_quad=1800.

Phase 2 — self-consistent vs non-self-consistent cost

Mercury (Z=80), all-electron, corrected SPARC-atomSFE. Fixed: p=20, q=45, domain=13 Bohr, omega=4, polynomial mesh (concentration 2). Channels = Lmax + 4, since Hg has 4f occupied (l_occ_max = 3). Ratio is scRPA / RPA@DFT.

Cost ratio, measured and projected

N_fe	Lmax	n_quad	scRPA (s)	RPA@DFT (s)	ratio	scRPA MiB	RPA@DFT MiB	mem ratio	kind
10	30	450	17.8	6.6	2.70x	263.0	240.6	1.09x	measured
15	30	675	45.7	13.2	3.46x	374.7	333.0	1.13x	measured
25	30	1125	149.4	40.5	3.69x	650.1	586.0	1.11x	measured
40	5	1800	107.2	35.0	3.06x	1138.2	916.3	1.24x	measured
40	10	1800	177.7	54.0	3.29x	1153.6	966.5	1.19x	measured
40	20	1800	314.4	91.4	3.44x	1258.3	1093.9	1.15x	measured
40	30	1800	460.8	129.3	3.56x	1390.3	1193.9	1.16x	measured
150	80	6750	24921.2	5114.0	4.87x	17503.1	14961.2	1.17x	projected
200	100	9000	60538.1	11770.3	5.14x	30770.2	26239.6	1.17x	projected